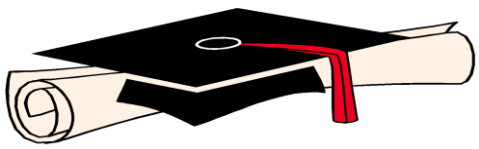
WAD

Web Application Development

Semester: 3	The Exam Code: 01
Duration: 90 minutes	Total Marks: 20



Practical Examination Paper

Practical Exam – ASP.NET Core MVC (Code First – GrabBike Management)

Requirements

● Use **Code First** to create a data model with **Entity Framework Core**
● Connect to a **SQL Server** database
● Use **Visual Studio 2019 or 2022**
● Use **Tag Helpers**
● The code must be **well-written, easy to understand, and fully commented**

Question 1: Managing GrabBike Trips and Drivers (12 points)

Data Model (3 points)

Create the following entity classes:

Driver.cs

● Id (int, primary key)
● FullName (string, **required**, min length 5, max length 100)
● PhoneNumber (string, **required**, max length 15)
● LicensePlate (string, **required**, max length 20)
● IsActive (bool)

Trip.cs

● Id (int, primary key)
● PickupLocation (string, **required**, max length 255)
● DropoffLocation (string, **required**, max length 255)
● DistanceKm (double, **required**, range 0.1 – 100)
● Fare (decimal, **required**, range 1000 – 1,000,000)
● TripDate (datetime)
● DriverId (int, foreign key)

Displaying a list of trips (2 points)

● Create an **Index.cshtml** view in the **Trips** folder
● Display the list of trips in a table
● Each trip should display:

- Pickup location
- Dropoff location
- Distance (km)

- Fare
- Trip date
- Driver name
  ● Use Tag Helper **asp-for** to display data

Adding a new trip (3 points)

● Create a **Create.cshtml** view in the **Trips** folder
● Display a form to enter trip information
● Use **asp-for** and **asp-validation-for** Tag Helpers
● Allow users to select a driver from a dropdown list
● Validate user input and display error messages
● Save the new trip to the database

Editing a trip (2 points)

● Create an **Edit.cshtml** view in the **Trips** folder
● Display a form to edit trip information
● Use **asp-for** and **asp-validation-for** Tag Helpers
● Allow users to change the assigned driver
● Validate input and update data in the database

Deleting a trip (2 points)

● Create a **Delete** action in the **TripsController**
● Ask the user to confirm before deleting
● Remove the trip from the database

Question 2: Pagination, Sorting, and Searching (8 points)

Paginating a list of trips (3 points)

● Implement pagination for the trip list
● Display **10 trips per page**
● Show a pagination bar
● Use **asp-action**, **asp-route-*** or custom logic

Sorting trips (2 points)

● Allow users to sort trips by:

- Trip date
- Fare
- Distance
  ● Use Tag Helpers or query string parameters for sorting

Filtering trips by driver (1 point)

● Allow users to filter trips by driver
● Send parameters through a **POST form**

Searching trips (2 points)

● Create a textbox for searching pickup or dropoff location
● Send parameters through a **POST form**
● Use **Where** clause to filter data
● Display search results

Notes

● You may add additional features to increase your score
● UI design is flexible
● The solution must follow **MVC pattern**
● Code must be **clear, structured, and commented**